Partial Fractions

ANSWER KEY

Section 1: Distinct Linear Factors

1 Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions $\frac{A}{x-1}+\frac{B}{x+2}$.

a. $A=\frac{4}{3}, B=\frac{11}{3}$ ✓

b. $A=4, B=11$

c. $A=1, B=4$

d. $A=2, B=3$

Cover-up rule: $A=\frac{5x-1}{x+2}\Big|_{x=1}=\frac{4}{3}$. $B=\frac{5x-1}{x-1}\Big|_{x=-2}=\frac{-11}{-3}=\frac{11}{3}$.

2 Find the partial fraction decomposition of $\frac{1}{x^2-9}$.

a. $\frac{1/6}{x-3}-\frac{1/6}{x+3}$ ✓

b. $\frac{1/3}{x-3}+\frac{1/3}{x+3}$

c. $\frac{1}{x-3}-\frac{1}{x+3}$

d. $\frac{1/6}{x-3}+\frac{1/6}{x+3}$

$\frac{1}{(x-3)(x+3)}=\frac{A}{x-3}+\frac{B}{x+3}$. $A=\frac{1}{x+3}\Big|_{x=3}=\frac{1}{6}$. $B=\frac{1}{x-3}\Big|_{x=-3}=-\frac{1}{6}$.

3 Evaluate $\int\frac{3}{(x+1)(x+4)}dx$.

a. $\ln\left|\frac{x+1}{x+4}\right|+C$ ✓

b. $3\ln|x+1|+3\ln|x+4|+C$

c. $\frac{1}{3}\ln\left|\frac{x+1}{x+4}\right|+C$

d. $\ln|(x+1)(x+4)|+C$

Decompose: $\frac{3}{(x+1)(x+4)}=\frac{1}{x+1}-\frac{1}{x+4}$. Integrating: $\ln|x+1|-\ln|x+4|+C=\ln\left|\frac{x+1}{x+4}\right|+C$.

Section 2: Repeated Linear Factors

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4 What is the correct template for decomposing $\frac{2x+7}{(x-2)^2(x+1)}$?

a. $\frac{A}{x-2}+\frac{B}{(x-2)^2}+\frac{C}{x+1}$ ✓

b. $\frac{A}{x-2}+\frac{B}{x+1}$

c. $\frac{Ax+B}{(x-2)^2}+\frac{C}{x+1}$

d. $\frac{A}{(x-2)^2}+\frac{B}{x+1}$

A repeated linear factor $(x-c)^2$ requires terms for each ascending power: $\frac{A}{x-c}+\frac{B}{(x-c)^2}$, plus terms for any distinct factors.

5 In $\frac{3x^2+1}{(x-1)^2}=A+\frac{B}{x-1}+\frac{C}{(x-1)^2}$, find C.

a. 4 ✓

b. 3

c. 1

d. 0

Multiply through by $(x-1)^2$: $3x^2+1=A(x-1)^2+B(x-1)+C$. Setting $x=1$ yields $3(1)^2+1=C\Rightarrow C=4$.

6 Decompose $\frac{x}{(x+1)^2}$.

a. $\frac{1}{x+1}-\frac{1}{(x+1)^2}$ ✓

b. $\frac{1}{x+1}+\frac{1}{(x+1)^2}$

c. $\frac{x}{x+1}-\frac{1}{(x+1)^2}$

d. $\frac{2}{x+1}-\frac{1}{(x+1)^2}$

Rewrite numerator: $x=(x+1)-1$. Then $\frac{(x+1)-1}{(x+1)^2}=\frac{1}{x+1}-\frac{1}{(x+1)^2}$.

Section 3: Irreducible Quadratic Factors

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7 What partial fraction form should be set up for $\frac{4x^2+2x+5}{(x-3)(x^2+4)}$?

a. $\frac{A}{x-3}+\frac{Bx+C}{x^2+4}$ ✓

b. $\frac{A}{x-3}+\frac{B}{x^2+4}$

c. $\frac{A}{x-3}+\frac{B}{x+2}+\frac{C}{x-2}$

d. $\frac{Ax+B}{x-3}+\frac{Cx+D}{x^2+4}$

An irreducible real quadratic factor (x^2+a^2) requires a linear numerator $(Bx+C)$ over the quadratic denominator.

8 In the decomposition $\frac{x^2+1}{(x-2)(x^2+1)}=\frac{A}{x-2}+\frac{Bx+C}{x^2+1}$, find A.

a. $\frac{5}{5}=1$ ✓

b. 0

c. 2

d. -1

Notice $\frac{x^2+1}{(x-2)(x^2+1)}=\frac{1}{x-2}$. Thus $A=1$ and $B=C=0$.

9 Evaluate $\int\frac{2x+3}{x^2+9}dx$.

a. $\ln(x^2+9)+\arctan\left(\frac{x}{3}\right)+C$ ✓

b. $\ln(x^2+9)+3\arctan(x)+C$

c. $\frac{1}{2}\ln(x^2+9)+\frac{1}{3}\arctan(x/3)+C$

d. $2\ln(x^2+9)+C$

Split into two integrals:
$\int\frac{2x}{x^2+9}dx+\int\frac{3}{x^2+9}dx=\ln(x^2+9)+3\cdot\frac{1}{3}\arctan\left(\frac{x}{3}\right)+C=\ln(x^2+9)+\arctan\left(\frac{x}{3}\right)+C.$

Section 4: Improper Rational Fractions (Long Division First)

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10 Before performing partial fractions on $\frac{x^3+2x^2+1}{x^2-1}$, what preliminary algebraic step is mandatory?

a. **Polynomial long division to separate into polynomial quotient plus proper remainder** ✓

b. Factoring the numerator completely

c. Applying the cover-up method directly

d. Multiplying numerator and denominator by x

Because the degree of the numerator (3) is greater than or equal to the degree of the denominator (2), long division is required first.

11 Perform polynomial long division on $\frac{x^3+2x^2+1}{x^2-1}$. What is the quotient?

a. $x+2$ ✓

b. $x-2$

c. x^2+2

d. x

$x(x^2-1)=x^3-x$. Subtract: $2x^2+x+1$. Next term: $2(x^2-1)=2x^2-2$. Subtract: $x+3$. The quotient is $x+2$ with remainder $x+3$.

12 Decompose the remainder term $\frac{x+3}{x^2-1}=\frac{x+3}{(x-1)(x+1)}$.

a. $\frac{2}{x-1}-\frac{1}{x+1}$ ✓

b. $\frac{1}{x-1}+\frac{2}{x+1}$

c. $\frac{3}{x-1}-\frac{2}{x+1}$

d. $\frac{1}{x-1}-\frac{1}{x+1}$

$A=\frac{x+3}{x+1}\Big|_{x=1}=\frac{4}{2}=2$. $B=\frac{x+3}{x-1}\Big|_{x=-1}=\frac{2}{-2}=-1$. Thus $\frac{2}{x-1}-\frac{1}{x+1}$.

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